JCP PAPER CAMPAIGN

# Research Update

Tunable coordinate-network bank with off-mesh quadrature · viscous Burgers 2D and 3D

23 October 2026

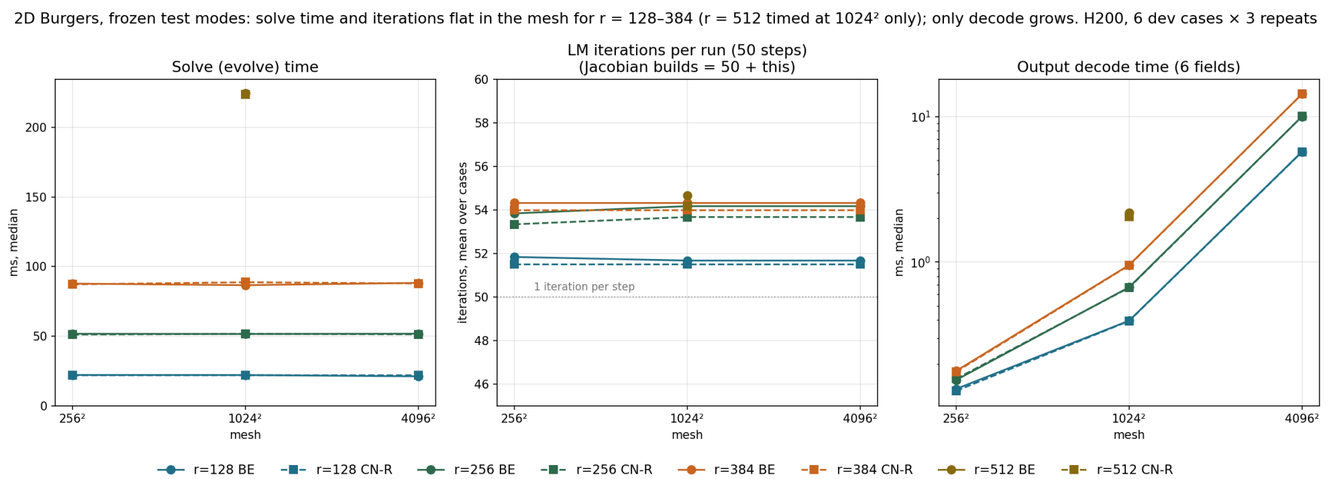
PART 1

# The dial and the mesh

One trained bank; choose how many shapes to use; the solve does not depend on the mesh.

QUESTION Do iterations or decoding depend on the mesh?

# Iterations are flat; only writing the output grows

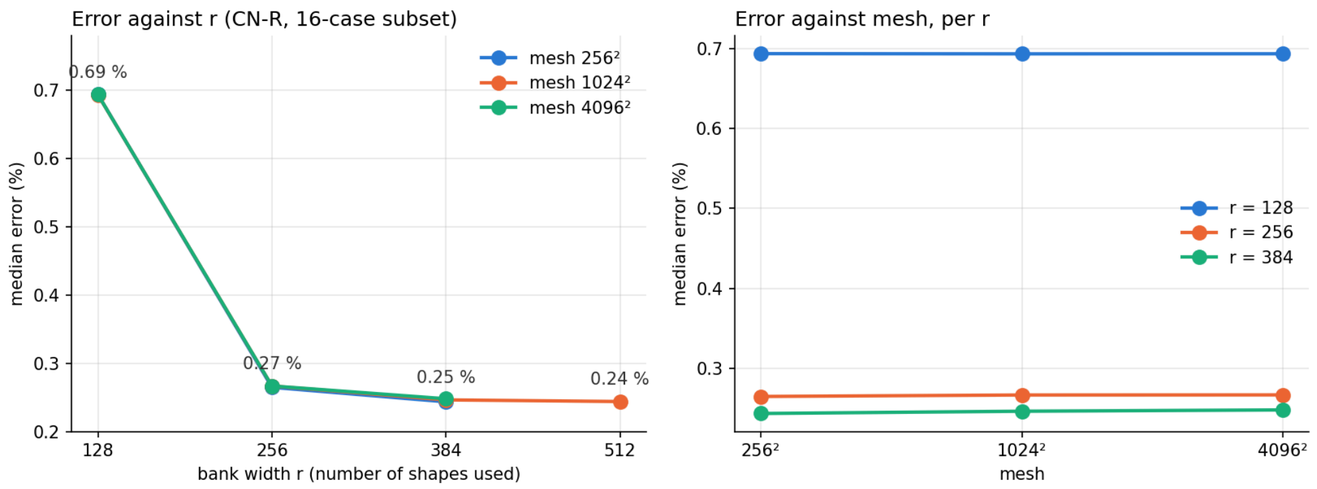



- **Solve:** r 128 ≈ 22 ms, r 256 ≈ 52 ms, r 384 ≈ 88 ms at every mesh
- **Iterations:** 51.5–54.3 Levenberg–Marquardt iterations per 50 steps, same on every mesh; at most 5 in any step
- **Decode:** 0.13–0.18 ms at 256² → 5.7–14.3 ms at 4096²
- Solid BE, dashed CN-R: schemes overlap

P1 timing job 5352971 (H200), frozen test modes, 6 dev cases × 3 repeats; Codex-checked · 5

QUESTION How do we get a dial r from one network?

# Reorder the 512 shapes so the first r are the best r

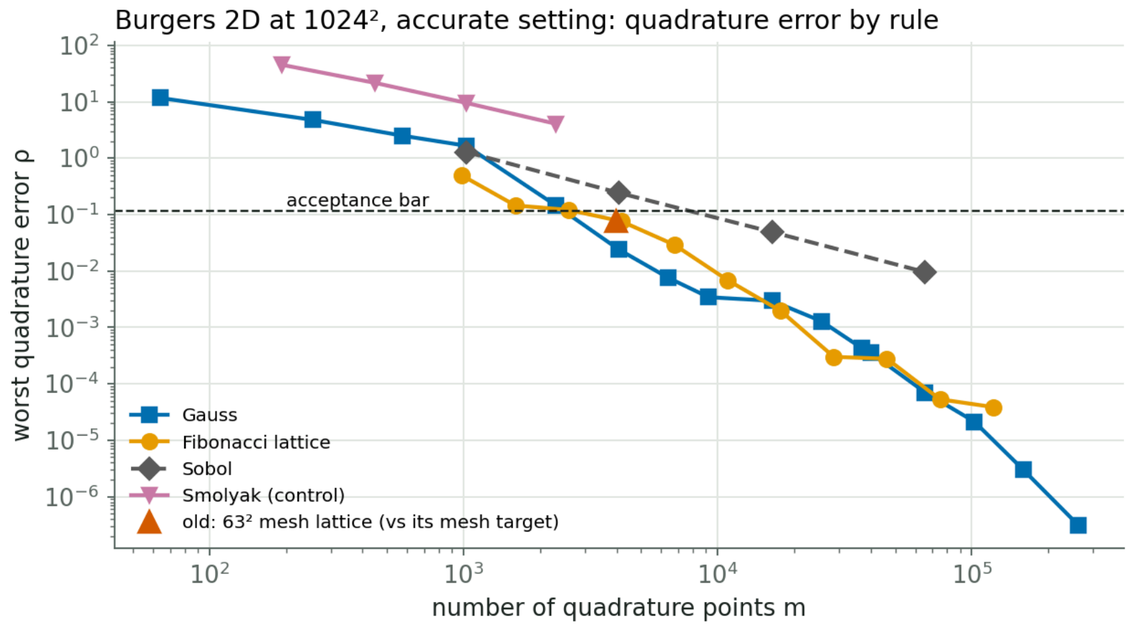



- QR of the shapes on a grid makes them orthonormal
- SVD of all training coefficients sorts directions by how much they are used
- Result: one rotation T; the first r columns are the best r-shape model
- Median error (CN-R, 16 cases): r 128 → 0.69 %, 256 → 0.27 %, 384 → 0.25 %, 512 → 0.24 %; flat after 256 (the time step takes over)

make_rotation.py; P1 analysis.json, 16-case subset (figs/make_error_vs_r_fig.py) · 3

QUESTION How many quadrature points does the nonlinear term need?

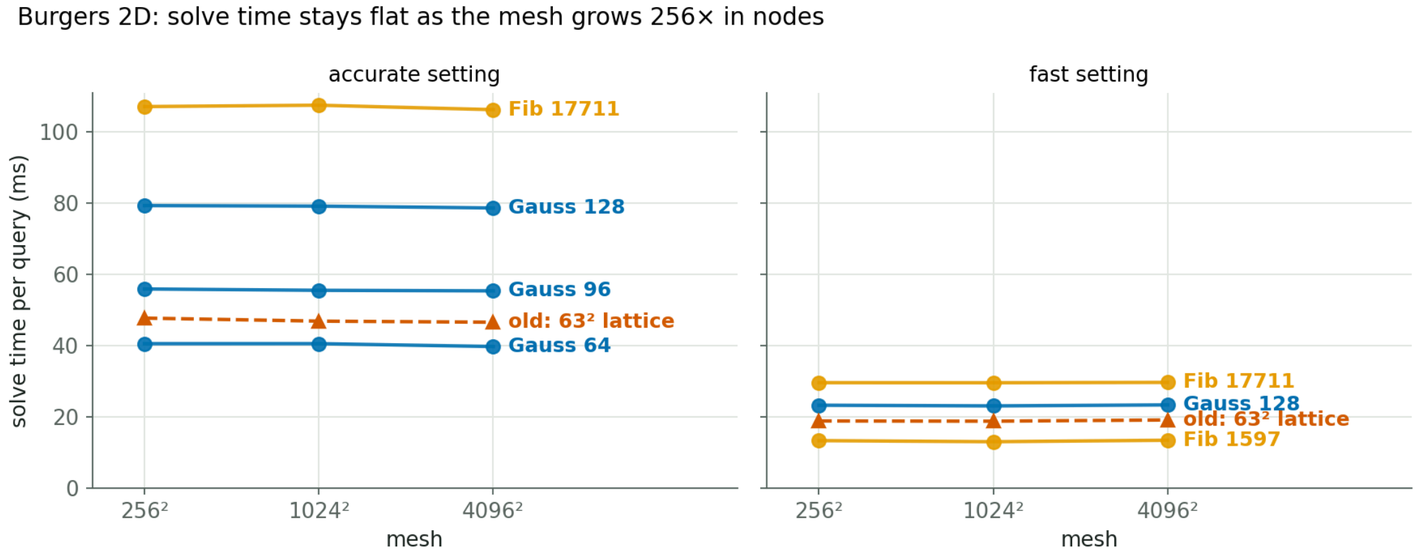
# Gauss and Fibonacci rules pass with a few thousand points; Sobol and Smolyak need far more




- ρ: worst relative error of the quadrature nonlinear term against a converged reference (Gauss 640²), over every state the model reaches
- Acceptance bar: ρ ≤ 0.116
- At about 4,000 points: Gauss 0.019, Fibonacci 0.033
- Sobol converges slowly; Smolyak (a control) never reaches the bar
- The old 63² mesh lattice (triangle) is measured against its own mesh target, so it is not directly comparable

QUESTION What do the quadrature points cost, and does that depend on the mesh?

# Cost grows with the number of points, never with the mesh




- **Accurate:** r = 384, 1536 tests; **fast:** r = 128, 512 tests
- Every rule is flat from 256² to 4096²: ratio 0.94–1.06 over all 40 measured arms
- More points cost more: Gauss 64² (4,096) → 96² (9,216) → 128² (16,384) → Fibonacci 17,711
- At 4096²: Gauss 96² 54.5 ms against the old 63² lattice 46.3 ms (accurate); Fibonacci 1597 13.3 ms against 18.7 ms (fast)

Off-mesh quadrature study 2D, one H200 job, dev6 × 3 repeats; solve = query − decode · 6

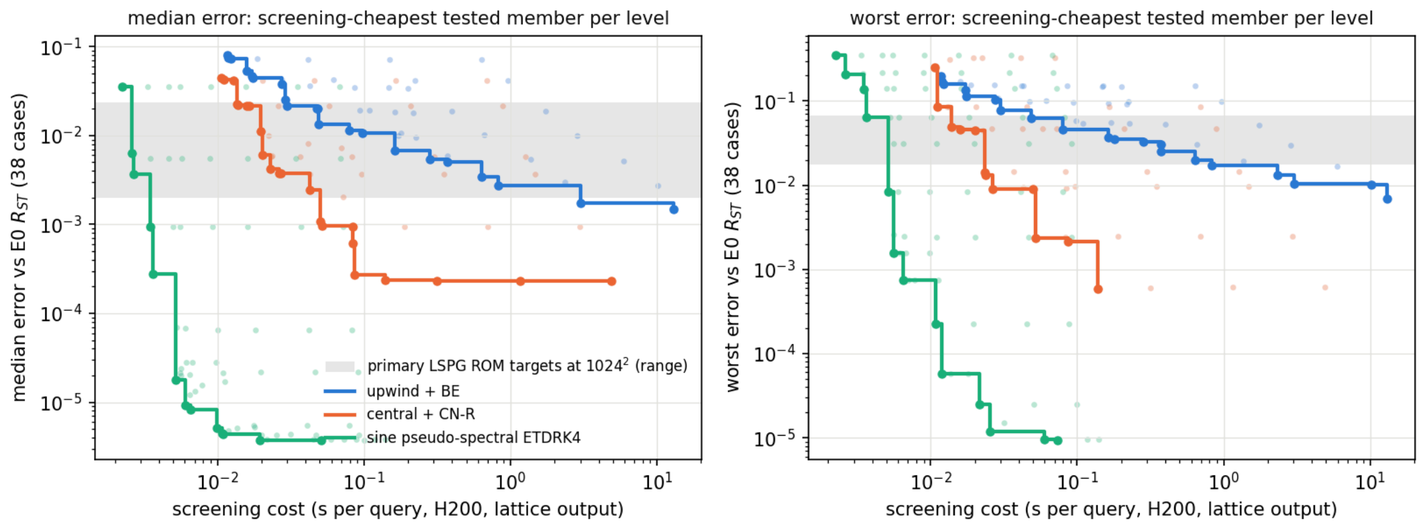
PART 2

# Accuracy against speed

How our model compares with full solvers and with tuned POD reduced models, in 2D and 3D.

QUESTION In 2D, how cheaply do full solvers reach our accuracy?

# A coarse spectral solver reaches our accuracy band at far lower cost

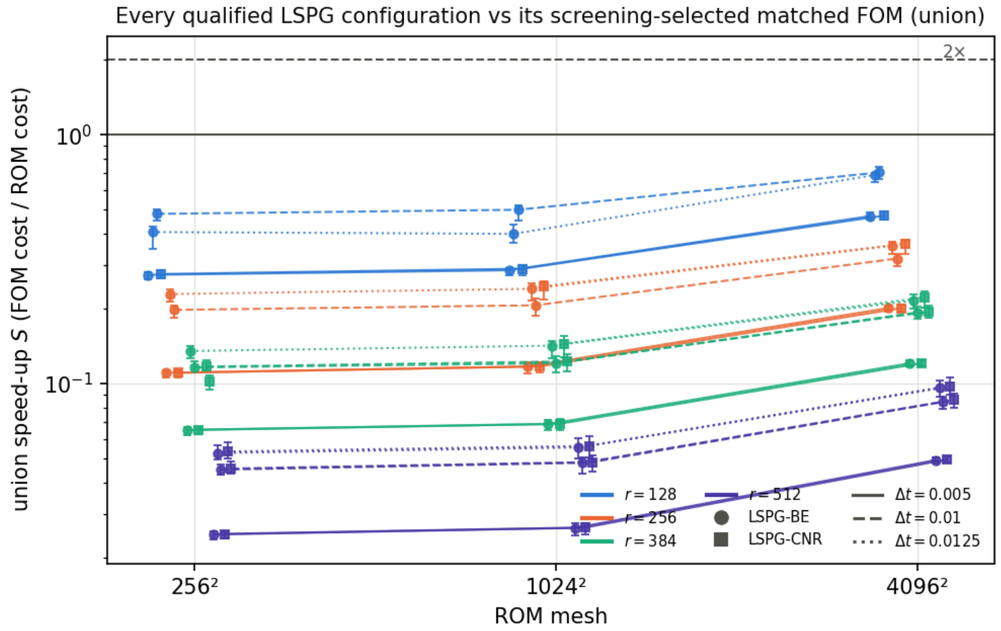



- Grey band: the accuracy our LSPG models reach at $1024^2$ (median and worst)
- Green: sine pseudo-spectral with ETDRK4 reaches that band at the lowest cost
- Orange (central + CN-R) and blue (upwind + BE) need much more time
- Example at $4096^2$: our r = 384 CN-R query 97 ms against 11.7 ms for the matched spectral solver

QUESTION In 2D, what is our speed-up at matched accuracy?

Slower

# Below 1× for every configuration, closing as the mesh grows

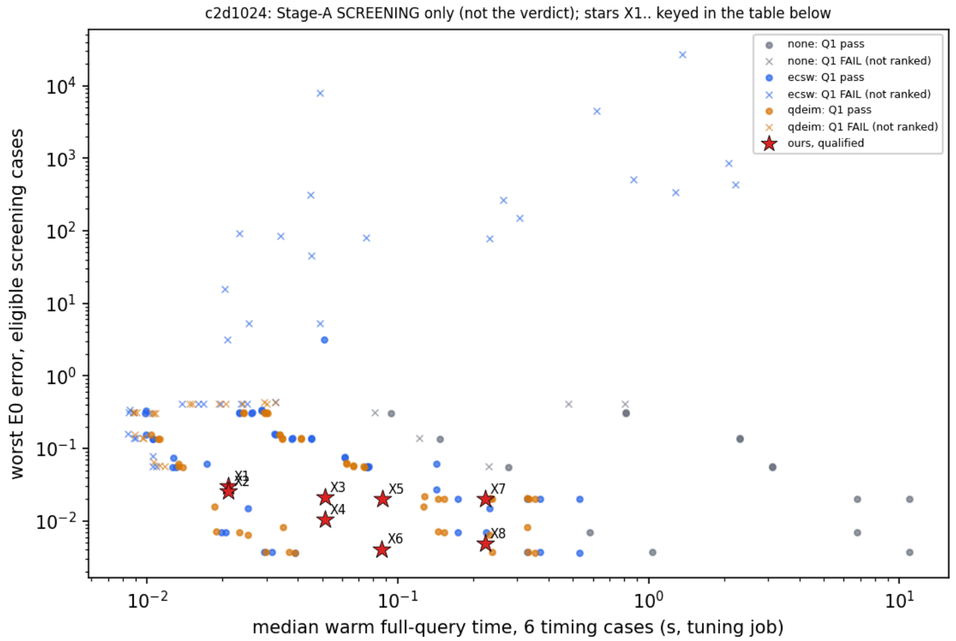



- Speed-up S = full-solver cost ÷ our cost at the same accuracy; above 1 means we are faster
- At $4096^2$: S = 0.049–0.71 across 21 configurations; none reaches the 2× bar
- S rises with the mesh because our solve cost is flat while the full solver's grows
- Smaller r (blue, r = 128) is closest to break-even

QUESTION In 2D, how do we compare with a tuned POD reduced model?

Slower

# POD with ECSW or Q-DEIM matches our accuracy in less time

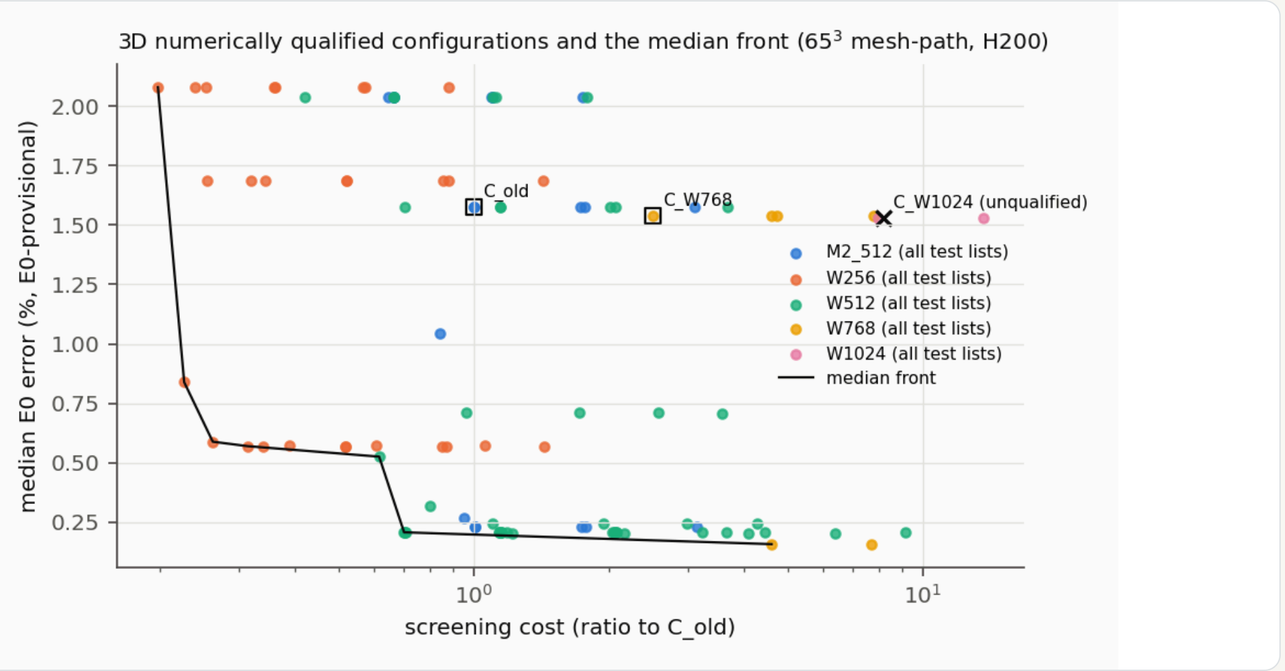



- Red stars: our qualified configurations; dots: POD + ECSW (blue), Q-DEIM (orange), none (grey)
- Dominated (faster and more accurate) in 8 of 8 configurations at $256^2$ and at $1024^2$
- Example: ours r = 384 CN-R 86.8 ms, worst 1.78 %; POD + ECSW 30.7 ms, worst 1.35 %

QUESTION In 3D, does the accuracy–cost dial work?

Works

# Yes: wider banks buy accuracy, and the qualified model is 2.95× faster than our earlier one

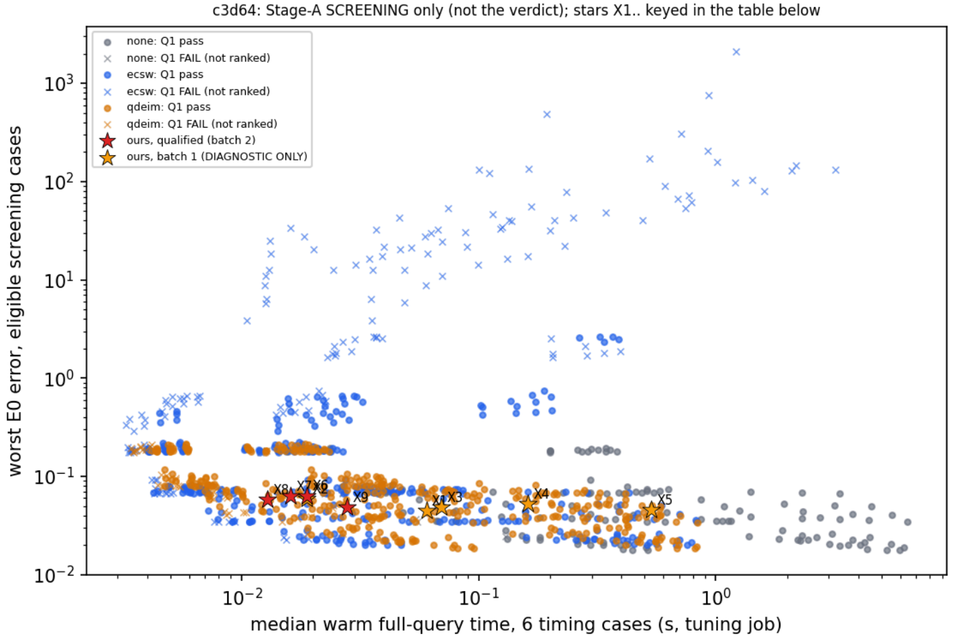



- Each dot is a numerically qualified 3D configuration (bank width, quadrature, solver)
- The black front: the cheapest configuration at each accuracy level
- Qualified speed-up 2.95× over our earlier wide model C_W768 at comparable accuracy
- Earlier unqualified claims (5.4×, 17×) are withdrawn

QUESTION In 3D, how do we compare with full solvers and POD?

Slower

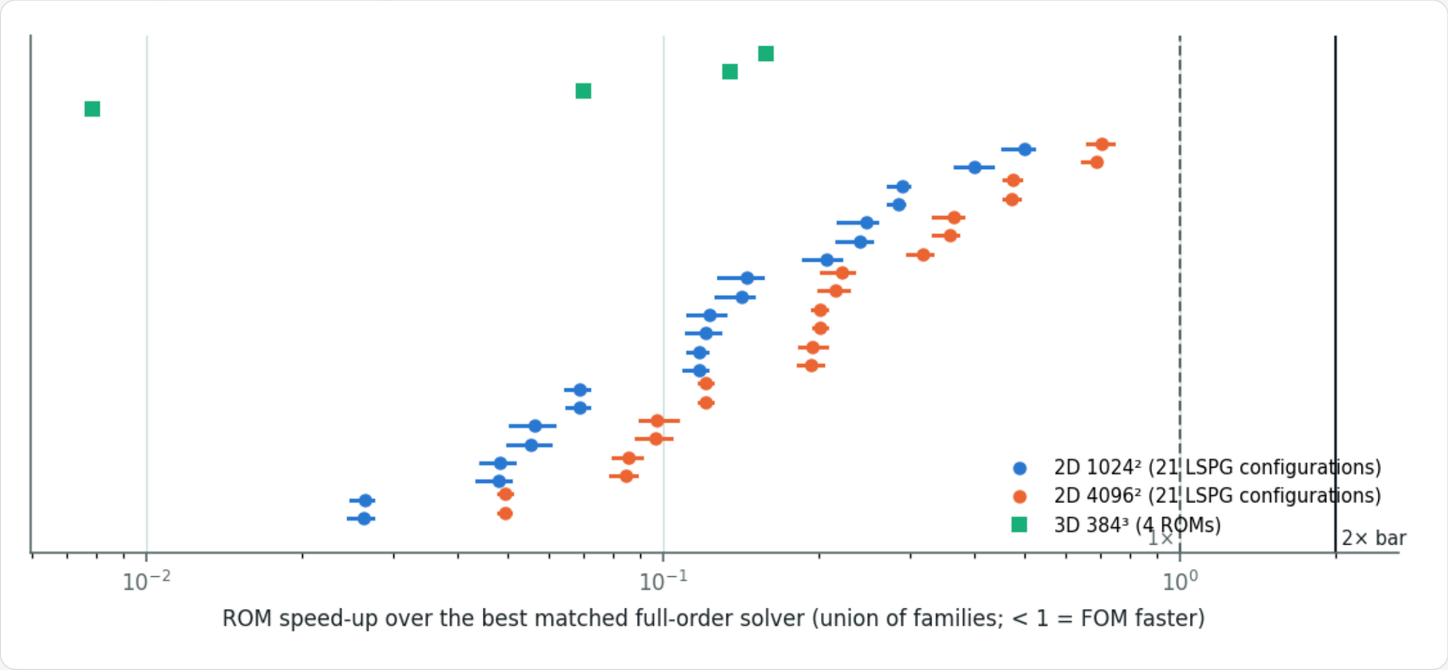
# Both are faster: the full solver by 6–130×, POD in most cells




- Against the best matched full solver at $384^3$: S = 0.0079–0.16 (4 models)
- Against POD at $65^3$: dominated in 5 of 5 (diagnostic batch) and 2 of 4 (qualified batch)
- POD here failed its own basis-adequacy check, so its results carry that restriction

QUESTION All together: where do we stand against the best full solver?

# Every configuration sits left of 1×; 3D is furthest behind

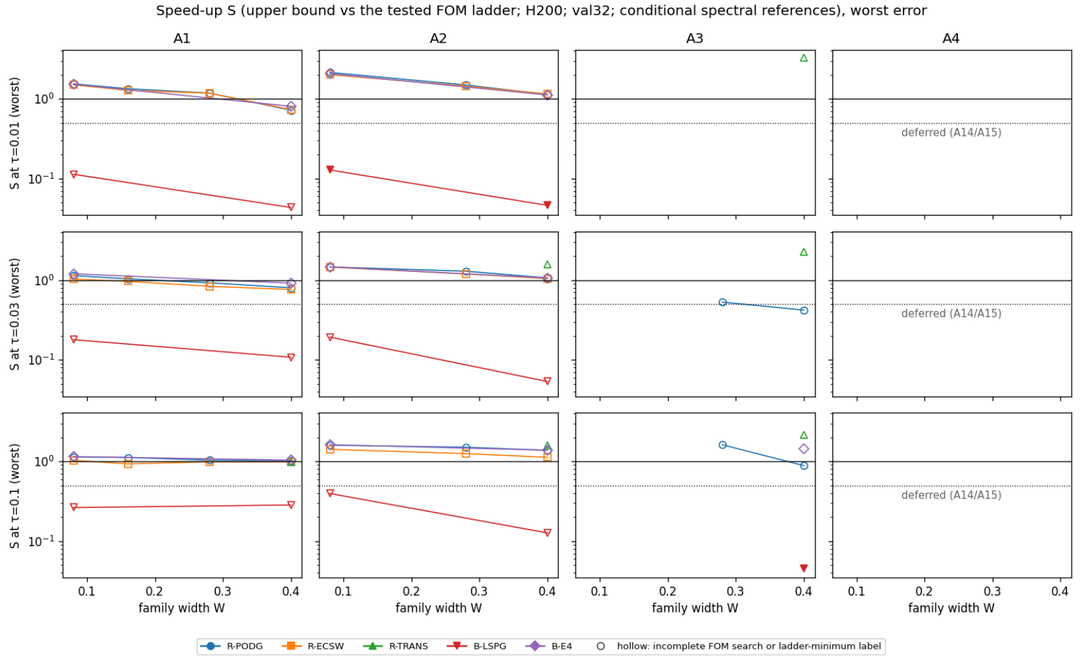



- Blue: 2D at $1024^2$; orange: 2D at $4096^2$; green: 3D at $384^3$
- Dashed line: break-even (1×); solid: the 2× bar we registered
- 2D gets closer at the bigger mesh; 3D stays at 0.008–0.16

QUESTION Newest: does a faster time stepper change the picture?

New

# With ETDRK4 the bank beats the tuned spectral solver by up to 2×, but so does POD




- Purple (our bank + ETDRK4): S = 0.81–2.07, above 1 in most cells
- Red (our bank, standard LSPG): S = 0.044–0.40
- Blue/orange (POD-Galerkin, POD + ECSW): up to 2.14 and 2.01, about the same as ours
- Best on narrow families (W = 0.08); falls as the family widens

QUESTION What does this add up to?

# Mesh-independent and tunable; not yet faster than the best alternatives

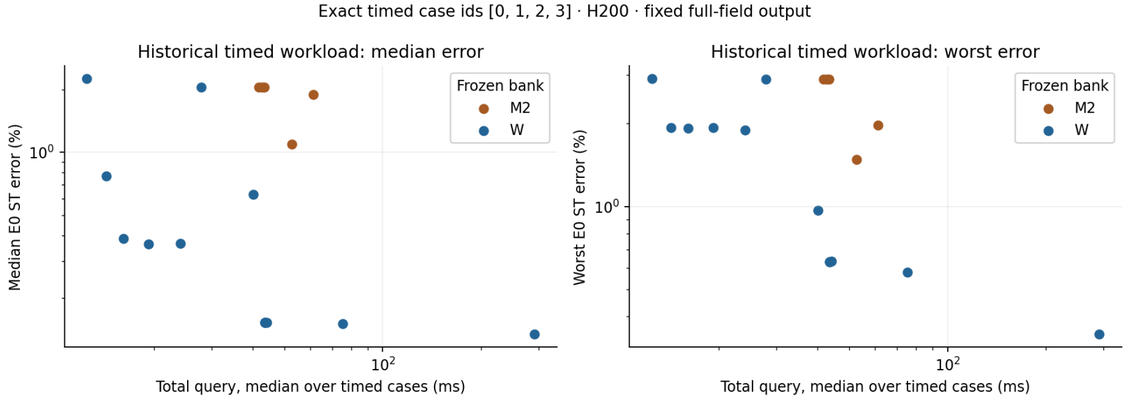
| Comparison | Where | Result |
|---|---|---|
| Mesh independence of cost and iterations | 2D and 3D | Yes: flat within 1.01–1.09 |
| Accuracy dial r | 2D 128→512; 3D | Improves to r ≈ 256–384, then flat |
| Against the best full solver (deployed LSPG) | 2D $4096^2$; 3D $384^3$ | Slower: S 0.049–0.71; 0.0079–0.16 |
| Against tuned POD + ECSW/Q-DEIM | 2D $256^2$, $1024^2$; 3D $65^3$ | Slower: dominated 16/16 (2D), 7/9 (3D) |
| Against our earlier 3D model | 3D $65^3$ | Faster: 2.95× (qualified) |
| Bank + ETDRK4 on the box (new) | 2D box, narrow to wide | 1–2× over spectral; level with POD (provisional) |

# The recovered frontier needs explicit populations

The strongest median and worst-case settings differ.

| Qualified historical role | Median | Worst |
|---|---|---|
| Best median | 0.158% | 7.17% |
| Best worst case | 0.205% | 3.14% |

The new inventory keeps references, exact cases, failed checks and later quadrature rounds separate.




The cost plot is re-scored on its exact saved timed cases [0, 1, 2, 3]. It is a subset view; full-validation accuracy is reported separately.

# A convergence repair with unchanged physical error

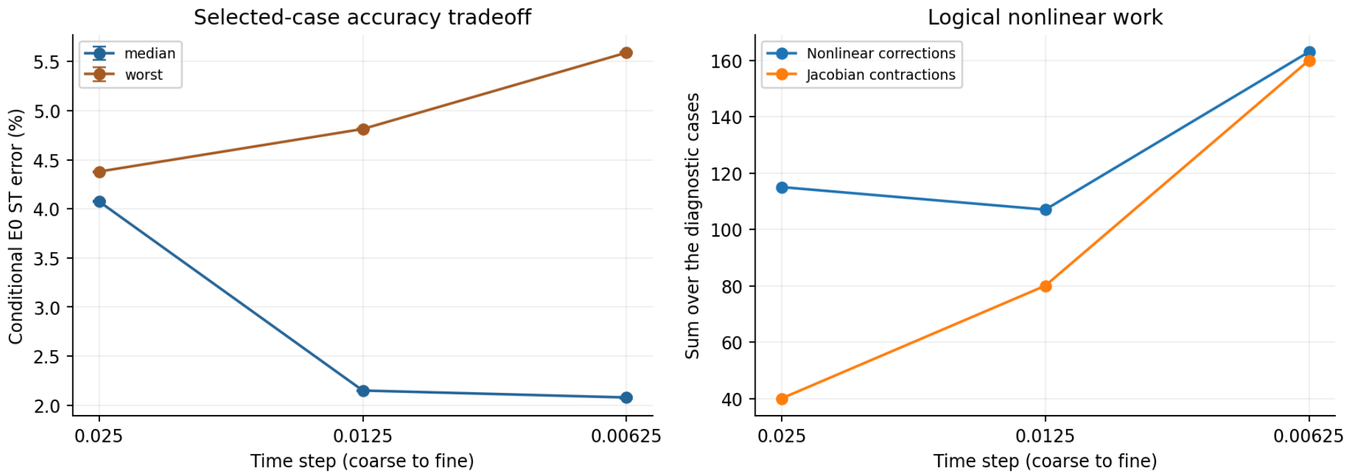
W1024 · validation case 30 · pilot job 6525183: a controlled replay repairs the nonlinear certificate while the worst physical error persists.

| Policy on the diagnostic case | Uncertified steps | Maximum E0 error |
|---|---|---|
| baseline | 1 | 3.9007% |
| extra | 0 | 3.9007% |
| refresh | 0 | 3.9007% |
| trust | 1 | 3.9007% |
| strict | 0 | 3.9007% |

Extra nonlinear work and more frequent matrix refresh address the demonstrated convergence gap. A larger trust radius alone does not. Fresh restarts are labelled separately from the original cached solver.

Selected difficult-case diagnostics. This table does not establish whole-population accuracy or speed gains.

# Time refinement improves the median and worsens outliers




Fixed bank/test/quadrature setting; case ids [0, 5, 30, 43].

On the fixed diagnostic cases, median error falls from 4.075% to 2.079%, while worst error rises from 4.379% to 5.59%.

Both endpoints pass the subset convergence checks. A lower median does not guarantee a better worst case.

The coarse and fine endpoints were selected after these diagnostics, before the full-validation sweep. The intermediate setting remains a valid measured tradeoff.

Error cancellation is a hypothesis. The inherited bank uses analytic advection and grid-dependent diffusion; it does not share POD's discrete residual.

# POD receives the same qualification scrutiny

Under the historical integration policy, the saved POD supports solve their sampled equations but fail the new complete-objective check.

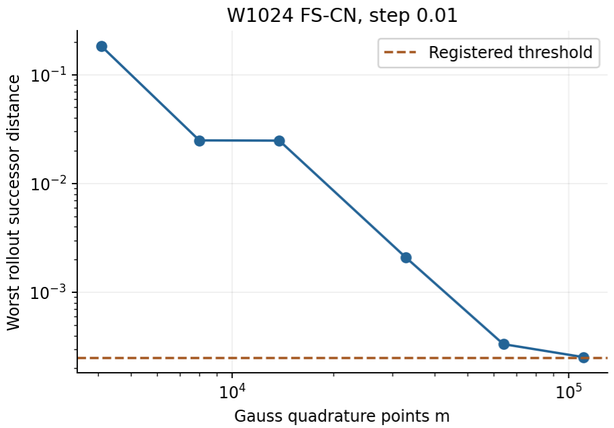
| Frozen support | Gradient discrepancy | Full-objective uncertified steps | Outcome |
|---|---|---|---|
| k8 | 0.015424 | 60/60 | Fails new contract |
| k16 | 0.0012332 | 6/60 | Fails new contract |

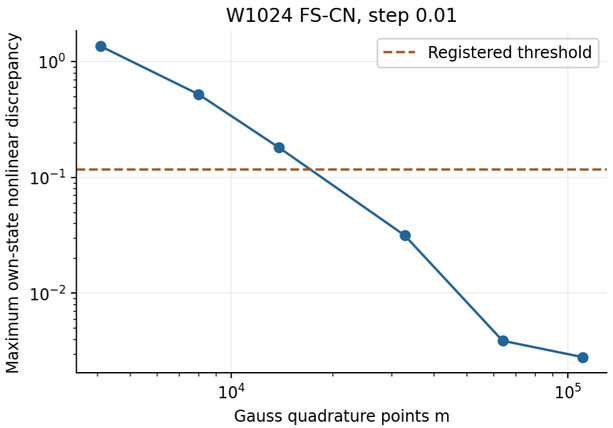
The historical 7/9 domination headline pooled 5 diagnostic and 4 qualified bank rows. Within the qualified subset: 2/4 dominated, 2 withheld. Old POD hyper-reduction convergence was not gated.

The single larger-support constructor stopped at 13673 of 16384 points and failed its optimality and banned-point checks. No larger-support rollout ran. Qualified unsampled POD remains available; the qualified HR comparison stays open. The diagnostic failures above use the new conservative rule and do not establish a bank advantage.

The existing k16 rule also fails the field-agreement check under tuned CN (maximum gap 0.00046528); its scalar reference-error changes remain within uncertainty. No full-validation HR release follows.

# Close missing evidence before extending the frontier




Completed quadrature trajectories are reused. The missing finer-rule check closes a specific evidence gap.

A qualified refined frontier needs whole-population solver and quadrature checks, same-invocation timing, and matched output requirements.

Reduced evolution, input preparation, output decoding, offline setup and memory remain separate costs.

No new general advantage over POD or the fastest tested full solver is established by the diagnostic pilots.